

Stage 1 — Dataset creation

Raw HC judgments
(5 domain buckets)

OpenNyAI
NER + RR + summarizer

LLM cross-validated
outcome labelling
(8-Q consistency prompt)

Multi-hearing
consolidation
(case-number merge)

Stage 2 — Graph building

Leakage-safe
preprocessing
(role drop, phrase mask)

Section assembly
(6 section texts)

Local case-star +
global authority graph

Node init: text emb
⊕ scalars (12 / 12 / 10)

Stage 3 — LJP (training & evaluation)

Dual text encoder
(BGE-M3 / InLegalBERT)

HGT backbone
(typed proj., type embed,
rel-aware multi-head attn.)

5-fold CV
per-bucket + cross-bucket

Ablation grid
(8 graph variants
± LR decay)

Stage 4 — Explainability & hearing-wise analysis

Frozen inference +
case embeddings

Heterogeneous
PGExplainer training

Importance ranking
(L -hop BFS, max edge score)

Evidence diagnostic +
embedding k -NN +
top- k sufficiency

Stage construction
(initial / accumulated)

Self-contained
test graph + padding

Ensembled inference +
transitions, what-changed